Chan and Lakonishok (1995); Keim and Madhavan (1997); Breen, Hodrick and Korajczyk (2002). Third, price impact results from information leakage and liquidity needs. Fourth, market conditions affect the underlying costs, e.g., Beebower and Priest (1980); Wagner and Edwards (1993); Perold and Sirri (1993); Copeland and Galai (1983); Huang and Stoll (1994), etc. Finally, trading strategy (style) influences trading cost, e.g., Kyle (1985); Bertismas and Lo (1998); Grinold and Kahn (1999) and Almgren and Chriss (1999, 2000). Following these results, we are finally ready to define the essential properties for a market impact model.

Essential Properties of a Market Impact Model

Based on these research and empirical findings, we postulate the essential properties of a market impact model below. These expand on those published in *Optimal Trading Strategies* (2003) and *Algorithmic Trading Strategies* (2006).

P1. *Impact costs increase with size*. Larger orders will incur a higher impact cost than smaller orders in the same stock and with the same strategy.

P2. *Impact costs increase with volatility*. Higher volatility stocks incur higher impact costs for the same number of shares than for lower volatility stocks. Volatility serves as a proxy for price elasticity.

P3. *Impact cost and timing risk depend on trading strategy* (e.g., trade schedule, participation rate, etc.). Trading at a faster rate will incur higher impact cost but less market risk. Trading at a slower rate will incur less impact but more market risk. This is known as the trader's dilemma. Traders need to balance the trade-off between impact cost and risk.

P4. *Impact costs are dependent upon market conditions and trading patterns*. As the order is transacted with more volume the expected impact cost will be lower. As the order is transacted with less volume the expected impact cost will be higher.

P5. *Impact cost consists of a temporary and a permanent component*. Temporary impact is the cost due to liquidity needs and permanent impact is the cost due to the information content of the trade. They each have a different effect on the cost of the trade.

P6. *Market impact cost is inversely dependent upon market capitalization*. Large cap stocks have lower impact cost and small cap stocks have higher impact cost in general (holding all other factors constant). Some difference in cost across market capitalization categories, however,